

Radio latency and correction-burst findings

race-tracker · XBee-PRO 900HP (S3B) shared radio · measurements of 9 October 2026 · one Mac, two radios on USB-serial adapters, RTK base running

Summary

- **A telemetry transmission conflicts with the correction burst only in a short window.** Frames written from about 40 ms before the burst start to about 10-20 ms after it were delayed by 20-35 ms (or, rarely, lost), and in that window 46% of correction bursts arrived incomplete (19 of 41), against 1.8% (2 of 114) otherwise. Outside it, latency is normal.
- **The burst itself is short and regular:** every 1000 ms, median 28 ms wide, 95th percentile 31 ms, widest 33 ms (focus run).
- **Latency is steady when clear of the burst:** 13.0 ms median for a 26-byte frame, with a 90th percentile of 13.3 ms. About 1-2% of frames take 25-35 ms longer than that, and one in 600 took 306 ms.
- **Delay grows predictably with frame size:** about 7.6 ms plus 0.21 ms per byte (26 bytes: 13.0 ms; 84 bytes: 25.1 ms).
- **Consequence for the schedule:** block telemetry writes from 40 ms before to 20 ms after the burst start, plus a guard - roughly 80 ms per burst, not the 180-276 ms assumed earlier. That leaves about 92% of each 1 s cycle (96% of a 2 s cycle) for telemetry.
- **The transmit gate and slot scheduler were then tested on the same radios (runs D-G below):** with the gate on, 0 of 300 frames were written in the conflict window and 0 of 87 bursts came through incomplete, against 8 of 23 (35%) bursts with a nearby frame when sending without it. Latency was unchanged (17 ms median for a 46-byte frame) and the longest hold was 83 ms.
- **Slots work as designed:** every frame was written within a few ms of its slot opening, with no overlap between boats. Two boats at 2 Hz and four boats at 5 Hz (84-byte batches, 40 ms slots) ran with 0 bursts incomplete, 0 drops and 0 expired frames. Frames wait up to about 1 s for their slot.
- **Not yet covered:** a Raspberry Pi's own delays, other adapter types, two real boats on separate radios, correction loss at a real rover, and the radio's on-air time per packet (estimated, not measured).

What was measured

`npm run radio-latency` opens two radios on the same machine. One sends frames at known times; the other decodes them. Both ends use the same clock, so the time from the app's `send()` to the other radio's decoded frame needs no clock synchronisation. The listener also timestamps each RTCM correction burst (the first RTCM message after a quiet gap), so every test frame can be placed relative to the burst. The delay includes the serial write, the radio's packetizing and RF time, the serial read and USB/Node scheduling - everything that makes a slot's start uncertain.

Run	Frame	Frames	Settings
A	26-byte position frame	300	default; sends spread over the whole second
B	84-byte batch frame (4 fixes)	600	LATENCY_BATCH=4 LATENCY_FRAMES=600
C (focus)	26-byte position frame	150	LATENCY_FOCUS=1: one frame per burst, swept from 80 ms before to 160 ms after the burst start

Results

	A: 26 B	B: 84 B	C: 26 B focus
Frames sent / received / lost	300 / 293 / 7 (2.3%)	600 / 589 / 11 (1.8%)	150 / 149 / 1 (0.7%)

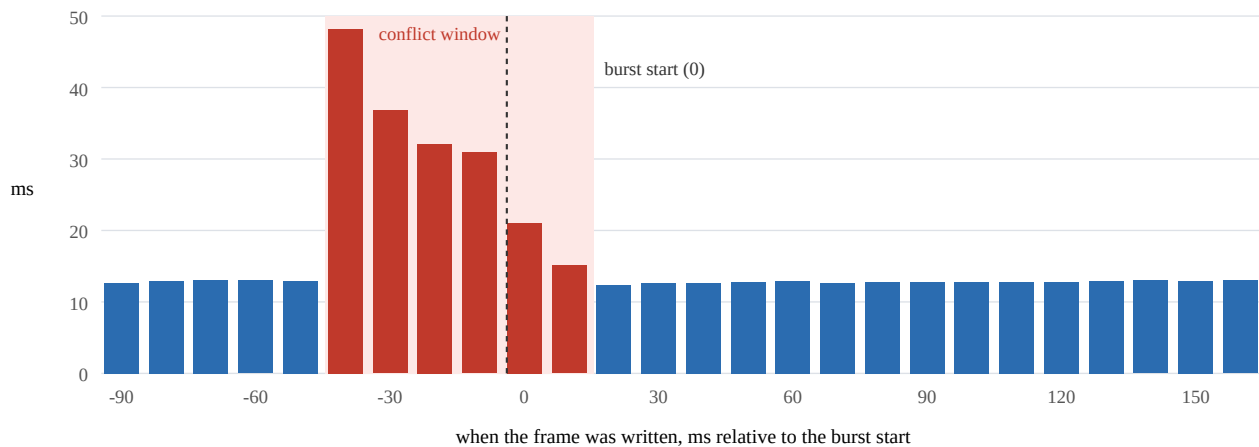
	A: 26 B	B: 84 B	C: 26 B focus
Latency min / p50 (ms)	12.3 / 13.0	24.6 / 25.1	12.3 / 12.9
Latency p90 / p99 / max (ms)	13.3 / 36.1 / 45.4	25.3 / 49.1 / 306.5	31.4 / 48.2 / 52.5
Frames clear of the burst: p50 / p90 / p99 (ms)	13.0 / 13.2 / 15.9	25.1 / 25.3 / 49.1	12.9 / 13.3 / 48.2*
Lost, near the burst	7 of 23	7 of 185**	1 of 49
Lost, clear of the burst	0 of 276	4 of 412**	0 of 101
Bursts incomplete: with a test frame near / without	4 of 18 (22%) / 3 of 66 (4.5%)	9 of 33 (27%) / 6 of 131 (4.6%)	19 of 41 (46%) / 2 of 114 (1.8%)
Burst width (ms)	up to 33	median not reported; widest 273	median 28, p95 31, widest 33

* Includes two frames sent 30-40 ms before a burst, which the run's classification called "clear"; the phase table below shows they were delayed. ** Run B classified "near the burst" using its widest burst (273 ms), which made the window about 300 ms wide, so its near/clear split is diluted. The script now uses fixed edges measured in run C (-45 to +25 ms around the burst start).

Delay against frame size

Between the 26-byte and 84-byte runs the median rose 12.1 ms for 58 more bytes: a fixed part of about **7.6 ms** plus about **0.21 ms per byte**. Two passes through a 115200-baud serial line (write and read) account for 0.17 ms per byte; the rest is about the radio's air time at 200 kb/s (0.04 ms per byte). For a batch frame the data reaches the far end about 25 ms after the write, and the schedule can subtract that offset. Only the variation needs to go into the guard.

Median latency (ms) by when the frame was written - run C



10 ms buckets, 2-8 frames each. Latency is about 13 ms except in the red buckets, where the radio held the frame back (31-48 ms) and one frame was lost (0 to 10 ms).

Implications for the schedule

- **Blocked window:** hold writes from 40 ms before to 20 ms after the burst start (the first RTCM message the radio decodes), then add a guard of about 10 ms each side: roughly **80 ms per burst**.
- **Anchor on the burst start, as heard by the same radio,** and re-anchor every cycle. Two radios hearing the same bursts agreed within 0-3 ms (earlier two-listener capture); the base's own burst-to-burst timing wanders by about 5 ms.
- **Frames are delayed more than lost** near the burst (1 of 49 lost in run C), but the corrections are the casualty: transmitting in the window is what leaves bursts incomplete.

• **Guard:** frames clear of the burst vary by about 3 ms (26 B) at the 99th percentile, with a heavier tail of 1-2% at 25-35 ms for big frames. A guard of 10 ms covers most; 25 ms covers more; the rare 300 ms outlier was probably a host stall.

Boats that fit in a slotted schedule (one slot per boat per cycle)

Cycle	Rate	Bytes / slot	Packets	Slot, guard 10 ms	Boats	Slot, guard 25 ms	Boats
1 s	2 Hz	46	1	20 ms	46	35 ms	26
1 s	5 Hz	110	2	30 ms	30	45 ms	20
1 s	10 Hz	214	3	43 ms	21	58 ms	15
2 s	2 Hz	84	1	21 ms	89	36 ms	52
2 s	5 Hz	214	3	43 ms	45	58 ms	33
2 s	10 Hz	420	5	67 ms	28	82 ms	23
4 s*	2 Hz	168	2	33 ms	117	48 ms	80
4 s*	5 Hz	420	5	67 ms	57	82 ms	46
4 s*	10 Hz	840	10	124 ms	31	139 ms	27

Model: 80 ms blocked per correction burst; a 2 s cycle means corrections every 2 s (one burst per cycle); *a 4 s telemetry cycle with corrections every 2 s has two bursts per cycle (160 ms blocked). Time on the air is estimated as 8 ms per radio packet plus 0.04 ms per byte (200 kb/s); the 8 ms is an upper bound from the measured fixed delay and has not been measured on the air. Boats are the number of slots in the cycle after the blocked time. The cost of a longer cycle is that a boat's newest position can be that old on arrival.

Testing the gate and slots on the real radios (runs D-G)

`LATENCY_SCHEDULE=gate|slots` runs the real `TxScheduler` instead of sending at random times. The sender radio hears the base's bursts itself, as a boat would (it waits for three bursts), and each virtual boat sends through its own scheduler. The listener radio records when each frame arrives and whether each correction burst came through complete. Same two radios and RTK base as above; 46-byte frames are 2-fix batches.

Run	Scheduler	Load	Frames written	In conflict window	Incomplete bursts
D	none (random phase)	1 boat, 46 B every ~250 ms	300	22 near a burst; 1 lost	8 of 23 with a frame near them (35%); 0 of 60 without
E	gate only	1 boat, 46 B every 250 ms	300	0	0 of 87
F	gate + slots (3, 7), 40 ms	2 boats, 46 B every 1 s (2 Hz)	120	0	0 of 67
G	gate + slots (0, 5, 10, 15), 40 ms	4 boats, 84 B every 0.8 s (5 Hz)	300	0	0 of 68

	D: no scheduler	E: gate	F: 2 boats, slots	G: 4 boats, slots
Latency after the write, p50 / p90 (ms)	17.3 / 17.8 (clear); 23.7 / 37.6 (near)	17 / 18	17 / 17	25 / 43
Held before the write	-	23 of 300 held, longest 83 ms	p50 about 0.55-0.6 s, max 0.97 s	p50 about 0.5 s, max 0.97 s
Written after burst start vs slot opening	-	gate only (30 ms and later)	slot 3: median 152 (150); slot 7: median 312 (310)	slots 0/5/10/15: medians 32/232/432/632 (30/230/430/630)
Dropped / expired / spilled	-	0 / 0 / 0	0 / 0 / 0	0 / 0 / 0

	D: no scheduler	E: gate	F: 2 boats, slots	G: 4 boats, slots
Frames from different boats within 25 ms	-	-	0	0

Slots open 30 ms after the burst start (20 ms burst tail plus 10 ms guard), then 40 ms apart. In G some frames are written 20-36 ms after the slot opens: a second frame goes out right behind the first in the same slot, which is also why its latency (p90 43 ms) is higher than for a single frame.

What the runs show

- **The gate removes the corruption.** Sending without it left 35% of the bursts it touched incomplete (22% to 46% across the earlier runs); the gate left none incomplete across 87 bursts, with the same latency and no lost frames. Only 23 of 300 frames were held at all, for 83 ms at most.
- **Slots keep boats apart as well as clear of the burst.** Frames from different boats never arrived within 25 ms of each other.
- **The slot cost is waiting.** A boat has one slot per cycle, so a position waits up to about 1 s (p90 about 0.9 s) for its slot. Gate-only holds at most about 80 ms.
- **Overload is handled safely.** A first slots run sent 4 two-fix frames per second per boat (8 fixes per second, more than a real boat produces) into 40 ms slots. About 3 frames went out per slot, the rest waited about 4 s, 15 per boat expired, and 50 frames spilled to later cycles. Nothing was transmitted outside the slot or in the conflict window (0 of 260; 0 bursts incomplete), and the app logged "slot too small" warnings. Raise the slot width or batch more fixes per frame if a boat's traffic does not fit.
- **Fit.** With corrections every 1 s, about 915 ms is usable, so 40 ms slots give 22 boats (30 ms slots give 30). Corrections every 2 s allow about 47 boats at 40 ms.

Limits of these runs: all virtual boats shared one sender radio, so this tests the scheduling and slot spacing but not two real radios each finding the burst start on their own clock; one Mac, not a Pi; and the effect on a real rover's RTK fix was not measured.

Caveats and next steps

- One Mac with two adapters. A Raspberry Pi adds its own serial/USB delay and variation; repeat the test on a Pi, and with the adapters that will be used.
- The conflict window is measured from the listener's decoded first correction message. A rover with a different adapter delay will see the same burst slightly earlier or later; the tracker should learn the offset from its own hearing.
- The rare long outliers (a 306 ms frame, a 273 ms burst) may be a host stall; not yet explained.
- Next: run the boat app on a Pi with SIMULATE_GPS=1 and TX_GATE=1 (add TX_SLOT_MODE=1, TX_SLOT_MS=40 and a TX_SLOT per boat for slots) with the Mac listener on the other radio; then two real boats in different slots; then a field test comparing a rover's RTK-fixed percentage and correction loss with the gate off and on.

Appendix: raw reports (runs D-H abridged to the lines that matter; A-C verbatim)

Run A - 26-byte frames, 300

Frames sent: 300 received: 293 lost: 7 (2.3%) frame size: 26 bytes

One-way latency in ms (send -> decoded at the listener):

all n= 293 min 12.3 p50 13.0 p90 13.3 p99 36.1 max 45.4
jitter (p99 - p50): 23.1 ms spread (max - min): 33.1 ms

RTCM bursts heard: 84, every 1000 ms, up to 33 ms wide, up to 6 messages

near burst n= 16 min 12.3 p50 12.9 p90 34.8 p99 36.1 max 36.1 lost 7 of 23
clear n= 276 min 12.3 p50 13.0 p90 13.2 p99 15.9 max 45.4 lost 0 of 276

Did our test frames hurt the corrections the listener was receiving?

bursts with a test frame sent in/near them: 18, of which incomplete: 4
bursts with none: 66, of which incomplete: 3

Suggested slot guard: about 11 ms (= jitter 3.0 + anchor disagreement 3 + margin 5;

based on the "clear" frames). Treat it as a starting point: one host, two adapters, 300 frames.

Typical delay before a frame is heard: p50 13.0 ms

Run B - 84-byte batch frames, 600

Frames sent: 600 received: 589 lost: 11 (1.8%) frame size: 84 bytes

One-way latency in ms (send -> decoded at the listener):

all n= 589 min 24.6 p50 25.1 p90 25.3 p99 49.1 max 306.5
jitter (p99 - p50): 24.0 ms spread (max - min): 281.9 ms

RTCM bursts heard: 164, every 1000 ms, up to 273 ms wide, up to 6 messages

near burst n= 178 min 24.6 p50 25.1 p90 25.4 p99 47.4 max 50.4 lost 7 of 185
clear n= 408 min 24.6 p50 25.1 p90 25.3 p99 49.1 max 306.5 lost 4 of 412

Did our test frames hurt the corrections the listener was receiving?

bursts with a test frame sent in/near them: 33, of which incomplete: 9
bursts with none: 131, of which incomplete: 6

Suggested slot guard: about 32 ms (= jitter 23.9 + anchor disagreement 3 + margin 5;

based on the "clear" frames). Treat it as a starting point: one host, two adapters, 600 frames.

Typical delay before a frame is heard: p50 25.1 ms

Run C - focus sweep, 26-byte frames, 150

Frames sent: 150 received: 149 lost: 1 (0.7%) frame size: 26 bytes

One-way latency in ms (send -> decoded at the listener):

all n= 149 min 12.3 p50 12.9 p90 31.4 p99 48.2 max 52.5
jitter (p99 - p50): 35.3 ms spread (max - min): 40.2 ms

RTCM bursts heard: 155, every 1000 ms, up to 6 messages; width median 28 ms, 95th percentile 31 ms, widest 33 ms

near burst n= 48 min 12.3 p50 18.8 p90 38.2 p99 41.7 max 41.7 lost 1 of 49
clear n= 101 min 12.4 p50 12.9 p90 13.3 p99 48.2 max 52.5 lost 0 of 101

By when the frame was SENT, relative to the nearest burst start (10 ms buckets):

from..to ms	n	lost	p50 ms
-90..-80	3	0	12.7
-80..-70	5	0	12.9
-70..-60	6	0	13.1
-60..-50	7	0	13.0
-50..-40	8	0	12.9
-40..-30	2	0	48.2
-30..-20	7	0	36.8
-20..-10	8	0	32.1
-10..0	6	0	30.9
0..10	4	1	21.0
10..20	8	0	15.1
20..30	6	0	12.4
30..40	4	0	12.6
40..50	7	0	12.6
50..60	5	0	12.8
60..70	7	0	12.9
70..80	6	0	12.7
80..90	6	0	12.8
90..100	5	0	12.8
100..110	6	0	12.8
110..120	7	0	12.8
120..130	4	0	12.8
130..140	6	0	12.9
140..150	7	0	13.0
150..160	5	0	12.9
160..170	5	0	13.0

Sends from -40 ms to 10 ms around a burst start were lost or delayed - hold telemetry out of that range, plus a guard.

Did our test frames hurt the corrections the listener was receiving?

bursts with a test frame sent in/near them: 41, of which incomplete: 19
bursts with none: 114, of which incomplete: 2

Suggested slot guard: about 44 ms (= jitter 35.3 + anchor disagreement 3 + margin 5;
based on the "clear" frames).

Run D - no scheduler, 46-byte (2-fix) frames, 300

Frames sent: 300 received: 299 lost: 1 (0.3%) frame size: 46 bytes

One-way latency in ms (send -> decoded at the listener):

all n= 299 min 16.6 p50 17.3 p90 17.8 p99 41.4 max 285.6
jitter (p99 - p50): 24.1 ms spread (max - min): 269.0 ms

RTCM bursts heard: 83, every 1000 ms, up to 6 messages; width median 26 ms, 95th percentile 28 ms, widest 263 ms

near burst n= 22 min 16.7 p50 23.7 p90 37.6 p99 41.6 max 41.6 lost 1 of 23
clear n= 275 min 16.6 p50 17.3 p90 17.7 p99 18.6 max 19.8 lost 0 of 275

Did our test frames hurt the corrections the listener was receiving?

bursts with a test frame sent in/near them: 23, of which incomplete: 8
bursts with none: 60, of which incomplete: 0

Suggested slot guard: about 10 ms (= jitter 1.2 + anchor disagreement 3 + margin 5).

Run E - gate only (LATENCY_SCHEDULE=gate), 300 frames

Mode: gate frames written: 300

Boat LAT01: submitted 300, written 300, lost 0

written ms after a burst start: min 30 median 447 max 944
held before the write (ms): p50 0 p90 0 max 83
latency after the write (ms): p50 17 p90 18 p99 18 max 24

Frames written inside the conflict window (-45..+25 ms around a burst start): 0 of 300

RTCM bursts heard by the listener: 87, every 1000 ms

bursts with a frame written in their conflict window: 0, incomplete: 0
bursts with none: 87, incomplete: 0

Scheduler (first boat): 276 sent at once, 23 held (longest 83 ms), 0 dropped, 0 expired, 0 slot spills; bursts tracked: 87

Run F - slots 3,7 at 40 ms, 2 boats, 46 B every 1 s, 60 frames each

Mode: slots (slot width 40 ms) frames written: 120

Boat LAT01 (slot 3): submitted 60, written 60, lost 0

written ms after a burst start: min 150 median 152 max 168 (slot opens at 150 ms)
held before the write (ms): p50 599 p90 903 max 961
latency after the write (ms): p50 17 p90 17 p99 18 max 18

Boat LAT02 (slot 7): submitted 60, written 60, lost 0

written ms after a burst start: min 309 median 312 max 325 (slot opens at 310 ms)
held before the write (ms): p50 552 p90 846 max 965
latency after the write (ms): p50 17 p90 17 p99 18 max 18

Frames written inside the conflict window (-45..+25 ms around a burst start): 0 of 120

Frames from different boats arriving within 25 ms of each other: 0

RTCM bursts heard by the listener: 67, every 1000 ms

bursts with a frame written in their conflict window: 0, incomplete: 0
bursts with none: 67, incomplete: 0

Scheduler (first boat): 3 sent at once, 56 held (longest 961 ms), 0 dropped, 0 expired, 0 slot spills; bursts tracked: 67

Run G - slots 0,5,10,15 at 40 ms, 4 boats, 84 B every 0.8 s, 75 frames each

Mode: slots (slot width 40 ms) frames written: 300

Boat LAT01 (slot 0): written 75/75, lost 0; written after burst start: min 30 median 32 max 66 (opens at 30); held p50 464 p90 871 max 958; late

Boat LAT02 (slot 5): written 75/75, lost 0; written after burst start: min 230 median 232 max 252 (opens at 230); held p50 489 p90 868 max 954;

Boat LAT03 (slot 10): written 75/75, lost 0; written after burst start: min 430 median 432 max 467 (opens at 430); held p50 496 p90 891 max 966;

Boat LAT04 (slot 15): written 75/75, lost 0; written after burst start: min 630 median 632 max 654 (opens at 630); held p50 492 p90 882 max 949;

Frames written inside the conflict window (-45..+25 ms around a burst start): 0 of 300

Frames from different boats arriving within 25 ms of each other: 0

RTCM bursts heard by the listener: 68, every 1000 ms

bursts with a frame written in their conflict window: 0, incomplete: 0
bursts with none: 68, incomplete: 0

Scheduler (first boat): 4 sent at once, 71 held (longest 958 ms), 0 dropped, 0 expired, 0 slot spills; bursts tracked: 68

Run H - overloaded slots 3,7 at 40 ms, 46 B every 250 ms (8 fixes/s per boat), 150 frames each

Mode: slots (slot width 40 ms) frames written: 260

[txgate] WARNING: slot 3 (40 ms) is too small for this boat's traffic - frames are waiting for later cycles.

Boat LAT01 (slot 3): submitted 150, written 130, lost 0

written ms after a burst start: min 150 median 152 max 156 (slot opens at 150 ms)
held before the write (ms): p50 4245 p90 4838 max 4988
latency after the write (ms): p50 36 p90 36 p99 37 max 37

Boat LAT02 (slot 7): submitted 150, written 130, lost 0

written ms after a burst start: min 310 median 311 max 313 (slot opens at 310 ms)
held before the write (ms): p50 4207 p90 4798 max 4977

Frames written inside the conflict window (-45..+25 ms around a burst start): 0 of 260

Frames from different boats arriving within 25 ms of each other: 0

RTCM bursts heard by the listener: 47; with a frame in the conflict window: 0, incomplete: 0

Scheduler (first boat): 0 sent at once, 130 held (longest 4988 ms), 0 dropped, 15 expired, 50 slot spills